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Ubuntu
jakka_bhanu@LAPTOP-DCEVSCQ0: $ mkdir ~/practical4
jakka_bhanu@LAPTOP-DCEVSCQ0: $ cd ~/practical4
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ nano multi_wait.c
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ gcc multi_wait.c -o multi_wait
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ ./multi_wait
Parent PID = 1142
[PARENT] Created child 0 with PID = 1143
[PARENT] Created child 1 with PID = 1144
[CHILD 1] Started. PID = 1144, PPID = 1142. Will sleep 4 sec.
[PARENT] Created child 2 with PID = 1145
[CHILD 0] Started. PID = 1143, PPID = 1142. Will sleep 2 sec.

[PARENT] Now waiting for all children using wait() (order not guaranteed)...
[CHILD 2] Started. PID = 1145, PPID = 1142. Will sleep 6 sec.
[CHILD 0] Finishing now. PID = 1143
[PARENT] wait() returned. Child PID 1143 finished with exit code 100
[CHILD 1] Finishing now. PID = 1144
[PARENT] wait() returned. Child PID 1144 finished with exit code 101
[CHILD 2] Finishing now. PID = 1145
[PARENT] wait() returned. Child PID 1145 finished with exit code 102
[PARENT] All children reaped. Exiting.
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ nano multi_waitpid.c
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ gcc multi_waitpid.c -o multi_waitpid
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ ./multi_waitpid
Parent PID = 1189
[PARENT] Created child 0 with PID = 1190
[CHILD 0] Started. PID = 1190, PPID = 1189. Will sleep 6 sec.
[PARENT] Created child 1 with PID = 1191
[CHILD 1] Started. PID = 1191, PPID = 1189. Will sleep 4 sec.
[PARENT] Created child 2 with PID = 1192

[PARENT] Now waiting using waitpid() - waiting in creation order (0,1,2) on purpose...
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[PARENT] Now waiting using waitpid() - waiting in creation order (0,1,2) on purpose...
[PARENT] Specifically waiting for child 0 (PID 1190)...
[CHILD 2] Started. PID = 1192, PPID = 1189. Will sleep 2 sec.
[CHILD 2] Finishing now. PID = 1192
[CHILD 1] Finishing now. PID = 1191
[CHILD 0] Finishing now. PID = 1190
[PARENT] waitpid() returned. Child PID 1190 finished with exit code 100
[PARENT] Specifically waiting for child 1 (PID 1191)...
[PARENT] waitpid() returned. Child PID 1191 finished with exit code 101
[PARENT] Specifically waiting for child 2 (PID 1192)...
[PARENT] waitpid() returned. Child PID 1192 finished with exit code 102
[PARENT] All children reaped. Exiting.
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ nano zombie_demo.c
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ gcc zombie_demo.c -o zombie_demo
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ ./zombie_demo &
[1] 1230
Parent PID = 1230
[PARENT] Child PID = 1232 created. I will now sleep 30 seconds WITHOUT calling wait().
[CHILD] I am PID 1232. Exiting now.
[PARENT] The child has already exited, so it is now a ZOMBIE until I reap it.
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ ps -eo pid,ppid,stat,cmd | grep zombie
  1230    1059 S    ./zombie_demo
  1232    1230 Z    [zombie_demo] <defunct>
  1242    1059 S+   grep --color=auto zombie
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ [PARENT] Done sleeping. Now calling wait() to reap the zombie...
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ nano zombie_fixed.c
[1]+  Done                  ./zombie_demo
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ gcc zombie_fixed.c -o zombie_fixed
jakka_bhanu@LAPTOP-DCEVSCQ0: ~/practical4$ ./zombie_fixed & sleep 1 && ps -eo pid,ppid,stat,cmd | grep zombie_fixed
[1] 1286
Parent PID = 1286
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[PARENT] Specifically waiting for child 2 (PID 1192)...
[PARENT] waitpid() returned. Child PID 1192 finished with exit code 102
[PARENT] All children reaped. Exiting.
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ nano zombie_demo.c
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ gcc zombie_demo.c -o zombie_demo
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ ./zombie_demo &
[1] 1230
Parent PID = 1230
[PARENT] Child PID = 1232 created. I will now sleep 30 seconds WITHOUT calling wait().
[CHILD] I am PID 1232. Exiting now.
[PARENT] The child has already exited, so it is now a ZOMBIE until I reap it.
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ ps -eo pid,ppid,stat,cmd | grep zombie
  1230    1059 S    ./zombie_demo
  1232    1230 Z    [zombie_demo] <defunct>
  1242    1059 S+   grep --color=auto zombie
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ [PARENT] Done sleeping. Now calling wait() to reap the zombie...
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ nano zombie_fixed.c
[1]+  Done                  ./zombie_demo
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ gcc zombie_fixed.c -o zombie_fixed
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$ ./zombie_fixed & sleep 1 && ps -eo pid,ppid,stat,cmd | grep zombie_fixed
[1] 1286
Parent PID = 1286
[PARENT] Child PID = 1288 created. Sleeping 5 seconds...
[PARENT] NOTE: I never explicitly call wait() in this version -
[PARENT] the SIGCHLD handler reaps the child automatically and immediately.
[CHILD] I am PID 1288. Exiting now.
[SIGCHLD HANDLER] Reaped child PID 1288 immediately (exit code 0)
[PARENT] Done. Check the process table during those 5 seconds -
[PARENT] there should be NO zombie, unlike the previous program.
  1290    1059 S+   grep --color=auto zombie_fixed
[1]+  Done                  ./zombie_fixed
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical$

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